CRAZY FASHION • TECHNICAL SPECIFICATION

WEBSITE WORKFLOW & SYSTEM ARCHITECTURE

End-to-End Operational Manual & Code Flow

An in-depth, structured explanation of the CRAZY FASHION luxury streetwear e-commerce platform. Covers the complete user journey, catalog filtering engine, reactive shopping cart state, coupon mechanics, multi-step simulated payment gateway, and client-side persistence architecture.

CLIENT-SIDE ENGINE

Pure Vanilla ES6+ architecture with zero framework overhead. Instant DOM updates and dynamic event routing.

REACTIVE CART & COUPONS

Automated price calculations, tiered discounts, coupon validation (CRAZY50, WELCOME10), and free shipping thresholds.

CHECKOUT SIMULATION

Multi-step transactional pipeline supporting Cards, UPI QR scanner, and Cash on Delivery with invoice generation.

Application: CRAZY FASHION Web Platform
Tech Stack: HTML5 Semantic Markup, Vanilla CSS3, Modern ES6+ JavaScript
Data Model: Client State with LocalStorage & Cross-Tab Synchronization

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1. EXECUTIVE SUMMARY & ARCHITECTURE MAP

SECTION 01

CRAZY FASHION is a luxury streetwear e-commerce web platform engineered around the core philosophy of "*Where Premium Meets Affordable*". Built using high-performance vanilla web standards (HTML5, Vanilla CSS3, and ES6+ JavaScript), the application delivers an app-like single-page-application (SPA) experience while retaining the SEO and multi-route strengths of traditional multi-page documents.

1.1 High-Level Architectural Flow

The application cleanly decouples **Presentation (HTML/CSS)**, **State Management (JavaScript Object)**, and **Persistence (Browser LocalStorage)**:



1.2 Workspace Directory & File Responsibilities

FILE / PATH	TYPE	SYSTEM RESPONSIBILITY & ROLE
`index.html` / `Home.html`	Master Storefront	Primary landing page with Hero, Feature Strip, Catalog Grid, Drop Countdown, Video Lookbook, Customer Reviews, Concierge Form, and Modal dialogs.
`categories.html`	Filtered Route	Dedicated gallery for **Artisanal Shirts**. Initializes with `state.selectedCategory = 'shirts'` while retaining global cart and checkout modals.
`Nextpage.html`	Filtered Route	Dedicated gallery for **Denim & Trousers**. Initializes with `state.selectedCategory = 'jeans'` with full transactional capabilities.
`coming-soon.html`	Teaser & Pre-drop	VIP Vault Drop page with live countdown timer, 100-piece allocation status, capsule sneak peek, and early access waitlist forms.
`script.js`	Core Engine (53KB)	Unified controller containing catalog data, state container, search & filter logic, cart computations, promo engine, simulated checkout, and Web Audio API chimes.
`style.css`	Design System (75KB)	CSS custom properties system (Colors, Glassmorphism, Breakpoints, Animations, Drawer slide-outs, Radar scanner, and Print Invoice rules).
`ads.mp4`	Media Asset	Cinematic lookbook video campaign looping seamlessly with audio mute/unmute control.

> **System Design Note: Zero-Dependency Architecture**
> The entire application functions natively in modern web browsers without requiring bundlers, node servers, or external framework runtimes. This guarantees sub-millisecond interaction speeds and offline-compatible client storage.

2. END-TO-END USER JOURNEY WORKFLOW

SECTION 02

The customer journey within CRAZY FASHION follows a friction-free 7-phase e-commerce funnel designed to guide visitors from product discovery to secure simulated order placement:

1. **Discovery & Announcement Awareness**
   The visitor lands on `index.html`. The top marquee bar flashes promo code **CRAZY50** (50% OFF) and displays a synchronized countdown clock to the upcoming capsule drop. The Hero section highlights brand value metrics (15K+ Stylists, 4.9★ rating, 100% Pure Cotton) alongside direct CTA links.
   Marquee Bar • Hero CTA • Live Drop Clock

2. **Interactive Catalog Exploration & Real-Time Search**
   Users browse products using three concurrent mechanisms: (a) Category Pills (All, Shirts, Denim, T-Shirts), (b) Search input with instant autocomplete suggestions and live count badges, or (c) Sorting by Price (Low/High) and Top Customer Rated. The DOM updates dynamically without full-page reloads.
   DOM Filter Pipeline • Search Autocomplete • Live Badge Counters

3. **Garment Inspection, Sizing & Wishlisting**
   Each product card exposes active size buttons (S, M, L, XL or 30, 32, 34, 36). Clicking "Quick View" opens a modal with high-resolution imagery, MRP savings percentages, and garment fabric descriptions. Clicking the heart icon toggles the item in the user's Wishlist, immediately updating the top navigation counter badge.
   Quick View Modal • Size Selector • Reactive Wishlist Toggle

4. **Cart Aggregation & Audio Feedback**
   Clicking "Add to Cart" pushes the selected product ID and chosen size into the cart array. The Web Audio API synthesizes an affirmative high-frequency chime (D5 to A5), a floating toast appears, and the top bag counter animates. Clicking "Buy Now" adds the product and immediately slides out the cart drawer.
   Web Audio Chime • Slide-Out Drawer • Toast Notification

5. **Cart Adjustments, Coupon Validation & Shipping Calculation**
   Inside the slide-out drawer, customers can increment/decrement quantities or remove items. Entering coupons like **CRAZY50** deducts 50% from the subtotal. Orders above ₹999 automatically trigger Free Delivery (saving ₹99). The grand total recalculates instantly.
   Promo Engine (CRAZY50) • Threshold Free Shipping • Live Recalculation

6. **Checkout Modal & Multi-Payment Gateway Selection**
   Clicking "Proceed to Checkout" closes the drawer and displays the 3-step Secure Transaction modal. Customers input delivery information (Name, Phone, Address, PIN) and choose between: (1) Credit/Debit Cards, (2) UPI / QR Code instant scan, or (3) Cash on Delivery (COD).
   Form Validation • Dynamic Payment Sub-fields (Cards/UPI/COD)

7. **Transaction Radar Processing & Order Invoice Generation**
   Submitting the form activates a pulsing radar scanner screen simulating bank encryption (2.2s delay). Upon completion, a harmonic success chord (C5-E5-G5-C6) sounds, the cart clears, and a receipt displays with a randomized order ID (e.g., **#CF-849201**) and a "Print Invoice" trigger.
   Radar Animation • Order ID Generator • Printable Invoice Receipt

3. TECHNICAL STATE ENGINE & DATA PIPELINE

SECTION 03

The website operates on a centralized reactive state model defined in `script.js`. Every user interaction modifies the `state` object, which triggers targeted UI render cycles and persists changes to browser storage:

3.1 Central State Data Schema

```
const state = {
  cart: JSON.parse(localStorage.getItem("crazy_cart") || "[]"),        // [{ productId: 1, size: "M", qty: 2 }]
  wishlist: JSON.parse(localStorage.getItem("crazy_wishlist") || "[]"), // Product IDs: [1, 7, 10]
  selectedCategory: "all",     // 'all' | 'shirts' | 'jeans' | 'tshirts'
  searchQuery: "",             // Real-time typed keyword
  sortBy: "featured",          // 'featured' | 'price-asc' | 'price-desc' | 'rating'
  activeCoupon: null,          // null | { code: 'CRAZY50', discount: 0.50 }
  coupons: {
    CRAZY50: 0.50,             // 50% discount
    WELCOME10: 0.10,           // 10% discount
    FREESHIP: "free_shipping"  // Delivery fee waiver
  },
  selectedSizes: {}            // Map of productId -> chosen size string
};
```

3.2 LocalStorage Persistence Keys

STORAGE KEY	DATA TYPE	LIFECYCLE & PERSISTENCE BEHAVIOR
`crazy_cart`	JSON Array	Persists customer shopping cart items across page refreshes, tab closures, and multi-page routing (`index.html` ↔ `categories.html` ↔ `Nextpage.html`).
`crazy_wishlist`	JSON Array	Stores product IDs flagged as favorites by the user. Synced across all active browser tabs via the window `storage` listener.
`crazy_drop_target_time`	Unix Timestamp	Dynamic rolling launch date for the capsule drop countdown clock (maintains exactly 4 days, 18 hours from initial site entry).
`crazy_drop_waitlist`	JSON Array	Logs customer VIP waitlist entries (Name, Email, Target Drop Garment, Timestamp) directly into client storage.

3.3 Cross-Tab State Synchronization

If a customer opens the store in multiple browser tabs simultaneously (for example, browsing shirts on tab 1 while inspecting denims on tab 2), state remains unified in real-time through the window storage event listener:

```
window.addEventListener("storage", (e) => {
  if (e.key === "crazy_cart") {
    state.cart = JSON.parse(e.newValue || "[]");
    updateBadges(); renderCartDrawer();
  } else if (e.key === "crazy_wishlist") {
    state.wishlist = JSON.parse(e.newValue || "[]");
    updateBadges(); renderProducts();
  }
});
```

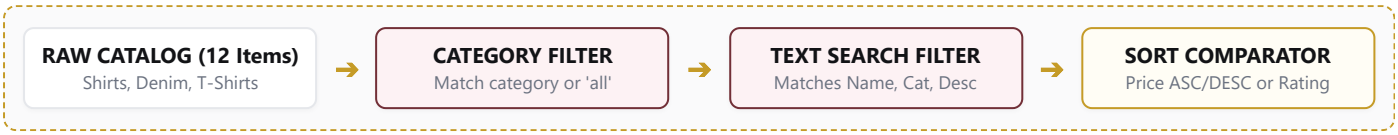
> **Performance Advantage**
> By isolating data mutations inside pure JavaScript functions and storing serializable state in LocalStorage, page transitions are instantaneous without requiring backend session re-validation.

4. FUNCTIONAL SUBSYSTEMS & CORE LOGIC

SECTION 04

4.1 Product Catalog & Filter Pipeline

The product engine maintains an array of 12 luxury garments. Filtering is executed through a functional pipeline:



4.2 Price Calculation & Free Shipping Formula

Every time the cart contents change, `getCartCalculations()` evaluates the financials:

```
function getCartCalculations() {
  let subtotal = state.cart.reduce((sum, item) => {
    const product = PRODUCTS.find(p => p.id === item.productId);
    return sum + (product ? product.price * item.qty : 0);
  }, 0);

  let discount = 0;
  if (state.activeCoupon && typeof state.activeCoupon.discount === "number") {
    discount = Math.round(subtotal * state.activeCoupon.discount);
  }

  // Orders > ₹999 get free delivery; else ₹99 fee
  const shipping = (subtotal > 999 || state.activeCoupon?.code === "FREESHIP" || subtotal === 0) ? 0 : 99;
  const grandTotal = Math.max(0, subtotal - discount + shipping);

  return { subtotal, discount, shipping, grandTotal };
}
```

4.3 Promo Code Matrix

COUPON CODE	DISCOUNT TYPE	BUSINESS CONDITION	EXAMPLE OUTPUT
CRAZY50	Flat 50% Off	Applies to entire cart subtotal	Subtotal ₹1,000 ➔ **-₹500 Discount**
WELCOME10	Flat 10% Off	Applies to new user orders	Subtotal ₹1,000 ➔ **-₹100 Discount**
FREESHIP	Shipping Waiver	Waives ₹99 delivery fee unconditionally	Delivery ₹99 ➔ **FREE (₹0)**
SECRET10	10% Vault Code	Granted upon VIP Capsule Drop registration	VIP early pass incentive

4.4 Web Audio API Chime Synthesizer

Instead of loading heavy MP3 audio files over the network, CRAZY FASHION uses the browser's native **Web Audio API** to synthesize pleasant haptic audio tones programmatically:

Cart Action Chime (Dual Tone)
Oscillates from D5 (587.33 Hz) to A5 (880.00 Hz) over 250ms with exponential gain fade-out to signal item addition.

Order Success Chime (Major Arpeggio)
Plays a 4-note ascending chord: C5 (523 Hz) ➔ E5 (659 Hz) ➔ G5 (783 Hz) ➔ C6 (1046 Hz) to celebrate completed purchases.

5. CHECKOUT SIMULATION & VIP DROP ENGINE

SECTION 05

5.1 Multi-Payment Simulated Transaction Flow

The checkout gateway in `index.html` and `categories.html` simulates an enterprise payment gateway experience with three distinct payment rails:

A **Credit / Debit Card Rail**
Provides input fields for 16-digit Card Number, MM/YY Expiry, and CVV with masked formatters. Pre-filled with a testing sandbox credential for instant user evaluation.

B **UPI / QR Code Rail**
Accepts Virtual Payment Addresses (VPA) and displays a simulated Dynamic QR Code for scanning via GPay, PhonePe, or Paytm with instant verification badges.

C **Cash on Delivery (COD) Rail**
Enables instant one-click door delivery verification without electronic payment requirements.

5.2 Transaction State Transition Diagram

STATE 1: FORM INPUT
Address & Payment Details

→

STATE 2: RADAR PROCESSING
2.2s Animated Pulsing Scanner

→

STATE 3: CONFIRMATION
Invoice, #CF-ID, Print Action

5.3 The 2026 VIP Vault Countdown System

Located on both `index.html#ComingSoon` and `coming-soon.html`, the countdown engine drives anticipation for limited-edition garment drops:

04d : 18h : 45m
SYNCHRONIZED PRECISION TIMER

86% RESERVED
14 VIP SPOTS REMAINING

UPCOMING DROP PIECE	CATEGORY	DROP PRICE	CRAFTSMANSHIP HIGHLIGHTS
Crazy Fashion Varsity Jacket	Outerwear	₹1,499 (MRP ₹2,499)	Heavy maroon wool, cowhide leather sleeves, brass snaps, serial-numbered tag.
Tokyo Monogram Resort Shirt	Artisanal Shirts	₹899 (MRP ₹1,299)	Silk-cotton blend, geometric monogram weave, mother-of-pearl buttons.
Japanese Raw Selvedge Wide Denim	Denim & Trousers	₹1,299 (MRP ₹1,999)	14.5oz shuttle-loom raw selvedge, red-line ID, architectural wide drape.

VIP Waitlist Mechanics
Clicking **"Notify Me"** on any capsule item opens the registration modal. Submitting the form saves the entry to `crazy_drop_waitlist` in LocalStorage, alters the card button to a checked "*VIP Waitlisted*" badge, and grants promo code **SECRET10**.

6. TESTING, VERIFICATION & FEATURE CHECKLIST

SECTION 06

To verify that all components of the CRAZY FASHION web platform are operating correctly, use the following operational checklist:

FEATURE MODULE	STEP-BY-STEP TEST PROCEDURE	EXPECTED SYSTEM RESPONSE
1. Category Filtering	Click "Artisanal Shirts" or select "Jeans" from dropdown or pills.	Grid displays only matching garments. Active pill highlights and counts sync.
2. Real-Time Search	Type "floral" or "denim" into header search input.	Autocomplete dropdown pops up with thumbnail links and matching counts.
3. Size Selection	Click different size chips ("XL" or "34") on any product card.	Selected size turns maroon/black with active state and binds to cart payload.
4. Quick View Modal	Click the eye icon overlay on any garment thumbnail.	Modal opens with detailed description, photo, and quick "Add to Cart" button.
5. Wishlist Storage	Click heart icon on any product card.	Heart fills red, badge count increments, and entry saves in `crazy_wishlist`.
6. Audio Chime & Cart	Click "Add to Cart" on any item.	Synthesizer plays D5-A5 chime, top bag counter bounces, and toast popup appears.
7. Promo Code CRAZY50	Open cart drawer, type `CRAZY50`, and click Apply.	A green discount line appears deducting 50% from the subtotal.
8. Free Delivery Rule	Add items until the subtotal exceeds ₹999.	Delivery fee flips from ₹99 to **FREE** automatically.
9. Radar & Invoice	Click "Proceed to Checkout", fill name/phone, and submit.	Pulsing radar activates for 2.2s, success chord plays, and printable receipt displays.
10. Cross-Page Routing	Navigate from `index.html` to `categories.html`.	Cart items and badges remain identical across pages via LocalStorage sync.

Summary & Project Conclusion

The CRAZY FASHION web platform successfully blends high-fashion visual aesthetics with modern, modular vanilla JavaScript engineering. By leveraging clean state machines, hardware-accelerated CSS animations, and native browser APIs, the application delivers a commercial-grade shopping experience that is fast, resilient, and effortless to navigate.